Referring to **T 39/82** (OJ 1982, 419) the board confirmed in **T 818/93** that in a **combination invention** all the features might be known per se – the invention resided in the way the features were interrelated, both structurally and functionally. In assessing the inventive step of the combination in question it was therefore of no consequence that a suitable structure was already known, provided its use and application in the conditions, and circumstances disclosed in the patent were not suggested by the cited prior art. For more on combination inventions see in this chapter I.D.9.3.

In **T 741/92** the invention involved the new use of a known means, namely a particular mesh structure. In the case of such inventions the board took the view that it was of little importance that the means was known per se if new properties and purposes came into play in its use. The known means was used in the invention to obtain a result not previously known or obvious.

Summing up in **T 301/90**, the board held that it was a generally accepted principle in the assessment of inventive step that, whereas the use of a known measure to achieve a known result on the basis of the expected inherent effect was not normally inventive, the indication of a new and non-obvious technical result, which could be achieved through these known effects (for application to the field of chemistry, see **T 4/83**, OJ 1983, 498 and to the field of physics, see **T 39/82**, OJ 1982, 419) might nevertheless convert the use of this known measure into a new and non-obvious tool for solving a new technical problem. It might thus represent an enrichment of the art and imply an inventive step (see **T 1096/92**, **T 238/93**).

In **T 590/90** the respondents argued that both the measures taken that distinguished the technical teaching of the contested patent from that of document 1 were already part of the prior art, and their application to the process described in document 1 was obvious. However, the board held that the application of a measure known as such, contrary to warnings given in several documents, was not obvious. Since this measure involved an inventive step, the overall process of claim 1 encompassing that measure likewise involved an inventive step: the modification of a known process by two measures, at least one of which was not obvious, rendered the entire process inventive.

9.14. Obvious new use

Although **T 59/87** had found that a claim to an inherent but hidden later use of a known substance could be novel, the subject-matter of such a claim would still lack inventive step if the prior art indicated a well-established link between the earlier and later uses (see also **T 544/94**).

In **T 112/92** (OJ 1994, 192) document (1), as the closest prior art, referred to the use of glucomannan as a thickener for an ungelled processed food product, but did not mention its function as a stabiliser. The board applied the principles set out in **T 59/87** (OJ 1991, 561) to the case in hand and stated that even if glucomannan did act as an emulsion stabiliser in preparing the product in accordance with document (1), this use would have been a hidden use. It came to the conclusion that the use of a substance as a stabiliser for emulsions, if not inextricably linked with its use as a thickening agent, was at least very closely related. The board held that it would have been obvious for the skilled person, knowing that glucomannan was effective as a thickening agent for emulsions, at least to try to find out if it was also effective as a stabiliser. Although **T 59/87** had found that a claim to an inherent but hidden later use of a known substance could be novel, the subject-matter of such a claim would still lack inventive step if the prior art indicated a well-established link between the earlier and later uses (see also **T 544/94**).

9.15. Need to improve properties

In its Headnote to **T 57/84** (OJ 1987, 53) the board stated: if a product is required to manifest a particular property (in this case a highly fungicidal effect) under various conditions, the superiority of the invention will depend on whether or not that property is **improved under all conditions** liable to be encountered in practice and particularly under the various conditions evolved in order to test it (in this case exposure to water and wind). If comparative tests are cited in support of that superiority, it is their combined results that have to be considered. The decisive factor is whether the invention outperforms the substance used for comparison in the tests **as a whole** (in this case, results in the need to use a significantly lower concentration of the pollutant substance), even if the substance used for comparison proves better in one of the tests.

Following **T 57/84**, it was stated in **T 254/86** (OJ 1989, 115) that an invention which relied on a substantial and surprising improvement of a particular property did not also need to show advantages over the prior art with regard to other properties relevant to its use, provided the latter were maintained at a reasonable level so that the improvement was not completely offset by disadvantages in other respects to an unacceptable degree or in a manner which contradicted the disclosure of the invention fundamentally (see also **T 155/85**, OJ 1988, 87). It was thus not necessary for there to be an improvement in every respect (**T 302/87**, **T 470/90**).

In **T 155/85** (OJ 1988, 87) it was further pointed out that subject-matter falling structurally between two particular embodiments of cited disclosure and displaying, in all relevant respects, effects substantially between those known for the same embodiments, lacked inventive step in the absence of other considerations.

9.16. Disclaimer

For information on disclaimers, see in particular chapter II.E.1.7. "Disclaimers".

9.17. Optimisation of parameters

In key decision **T 36/82** (OJ 1983, 269), the board stated that inventive step was not considered to be constituted by efforts directed at the concurrent optimisation of two parameters of a particular device by the simultaneous solution of two equations which were known per se and respectively expressed those parameters as functions of certain dimensions of the device. The fact that it had proved possible to find a range of values for the dimensions in question which provided an acceptable compromise between the two parameters could not be considered surprising where there were indications in the prior art suggesting that favourable results might be obtained by the method of calculation applied.

In a number of other decisions, all of which referred to **T 36/82** (OJ 1983, 269) the subject-matter was found not to involve an inventive step, particularly when the problem addressed was to find a suitable compromise between different parameters (**T 263/86**, **T 38/87**, **T 54/87**, **T 655/93**, **T 118/94**, **T 1861/17**). In **T 410/87** the board stated that it was part of the activities deemed normal for the skilled person to optimise a physical dimension in such a way as to reach an acceptable compromise, serving the intended purpose, between two effects which were contingent in opposing ways on this dimension (see also **T 409/90**, OJ 1993, 40; **T 660/91**; **T 218/96**; **T 395/96**; **T 660/00**, **T 1861/17**).

In **T 73/85** the board stated that in the case in hand the very fact that the problem of improving the property in question was solved not – as was normal – by means of a specific change in structural parameters, but by amending process parameters, had in fact to be considered surprising. In this case it did not matter that the individual reaction conditions claimed in the disputed patent were known per se; more important was whether the skilled person, in expectation of the sought-after optimisation, would have proposed the combination of measures, which were known per se, or – in the absence of possible predictions – would have tried to do so as a matter of priority.

In **T 500/89** the board established that the fact that individual parameter areas taken per se were known did not imply that it was obvious to combine them specifically to solve the problem according to the contested patent. The combination of the individual parameter areas was not the result of merely routine optimisation of the process according to document 1, as there was nothing in said document to suggest this combination.

9.18. Small improvement in commercially used process

In **T 38/84** (OJ 1984, 368) the board of appeal pointed out that the achievement of a numerically small improvement in a process commercially used on a large scale (here enhanced yield of 0.5%) represented a worthwhile technical problem which should not be disregarded in assessing the inventive step of its solution as claimed (see also **T 466/88**, **T 332/90**). In **T 155/85** (OJ 1988, 87) the board added that it was correct to say that even small improvements in yield or other industrial characteristics could mean a very relevant improvement in large-scale production, but the improvement had to be significant and therefore above margins of error and normal fluctuations in the field in consequence of other parameters. In **T 286/93** the invention related to a process for manufacturing wrapping paper and board. The results for the process had shown that the machine speed and the mechanical quality of the paper obtained had improved by some 3 % vis-à-vis a process in which the order in which aluminium polychloride and cationic starch were added had been reversed. Since a process of this kind was obviously intended for the production of paper on an industrial scale, even a small improvement had to be regarded as significant.

9.19. Analogy processes

The effect of a process manifests itself in the result, i.e. in the product in chemical cases, together with its internal characteristics and the consequences of its history of origin, e.g. quality, yield and economic value. It is well-established that analogy processes are patentable insofar as they provide a novel and inventive product. This is because all the features of the analogy process can only be derived from an effect which is as yet unknown and unsuspected (problem invention). If, on the other hand, the effect is wholly or partially known, e.g. the product is old or is a novel modification of an old structural part, the invention, i.e. the process or the intermediate therefore, should not merely consist of features which are already necessarily and readily derivable from the known part of the effect in an obvious manner having regard to the state of the art (**T 119/82**, OJ 1984, 217; see also **T 65/82**, OJ 1983, 327).

According to **T 2/83** (OJ 1984, 265), so-called analogy processes in chemistry are only claimable if the problem, i.e. the need to produce certain patentable products as their effect, is not yet within the state of the art.

In **T 1131/05** the board found that the conditions for granting a claim to an analogous process defined in **T 119/82** and **T 2/83** applied to the subject-matter of claim 10. It thus deemed a process claim directed to an analogy process to be new and inventive.

9.20. Envisageable products

In **T 595/90** (OJ 1994, 695) at issue was the inventiveness of a product which could be envisaged as such but for which no known method of manufacture existed. In its Headnote, the board noted that a product which could be envisaged as such with all characteristics determining its identity including its properties in use, i.e. an otherwise obvious entity, might nevertheless become non-obvious and claimable as such, if there was no known way or applicable (analogous) method in the art for making it and the claimed methods for its preparation were therefore the first to achieve this and do so in an inventive manner (see also **T 268/98**, **T 803/01**, **T 441/02**, **T 1175/14**).

In **T 268/98** the board found that the prior art did not comprise any technical information as to how to obtain the "pinning centers" according to claim 1. Therefore, even if the claimed product were to be considered highly desirable, there was no obvious method in the art to make it. Thus the average skilled person not even could, let alone would, have been able to arrive at the claimed product. The board, therefore, held that the product according to claim 1 involved an inventive step.

In **T 233/93** the combination of properties defining the claimed products had been a desideratum which the skilled community had striven to achieve. These properties, however had been considered to be irreconcilable. The board stated that such a desired product, which may appear obvious per se, may be considered non-obvious and be claimable as such, if there is no known method in the art to make it and the claimed methods for its preparation are the first to produce it and do so in an inventive manner (**T 1195/00**).

In **T 661/09** the board found the remaining features of claim 1 expressed no more than a set of desiderata, without any indication of a causal link between the desired properties and the constitution of the claimed device. Insofar as the claim did not define any concrete measures on how to ensure that the claimed properties were effectively obtained, the claimed properties remained at an abstract or conceptual level. The issue of inventive step boiled down to the question of whether or not the skilled person, in view of the available prior art and his/her common general knowledge, would in an obvious way have envisaged the claimed set of desiderata. The board found this question must be answered in the affirmative, concluding that claim 1 lacked an inventive step.

9.21. Examples of lack of inventive step

9.21.1 Foreseeable disadvantageous or technically non-functional modifications

In some decisions the subject-matter was found not to involve an inventive step, when the invention was the result of a foreseeable disadvantageous modification of the closest prior art (**T 119/82**, OJ 1984, 217; **T 155/85**, OJ 1988, 87; **T 939/92**, OJ 1996, 309; **T 72/95**; **T 694/13**).

The board in **T 119/82** (OJ 1984, 217) had already found that disadvantageous modifications did not involve an inventive step if the skilled person could clearly predict these disadvantages, if his assessment was correct and if these predictable disadvantages were not compensated by any unexpected technical advantage. More recently, the board in **T 2197/09** confirmed that inventive step cannot be acknowledged on the basis of a purely disadvantageous modification of the closest prior art.

9.21.2 Modifications to the closest prior art obvious to the skilled person

In **T 1862/15** the board accepted that an alternative means for achieving a known technical effect could, in some cases, be considered to involve an inventive step. However, achieving a known technical effect by means of modifications of the closest prior art which would be obvious to the skilled person could not be considered inventive.

9.21.3 Technical standards

In **T 519/12** the board held that it had to be expected of the skilled person that he would exercise his skills in the framework of technical standards in force in his field of activity. No inventive activity could be derived from a feature that simply reflected the contents of such a technical prescription.

9.21.4 Reversal of procedural steps

In **T 1/81** (OJ 1981, 439) the board held that in the absence of other features that from a technical point of view would contribute to patentability, the sequence in which the socket and pipe connection in the case in hand was made did not suffice to impart inventive step to the method claimed. The mere reversal of procedural steps in the production of component parts could not provide justification for inventive step.